

Temporal Swarm Federation: Phase-Coherent Multi-Agent Coordination via Oscillator Synchronization

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Abstract

We introduce *Temporal Swarm Federation* (TSF), a coordination paradigm for multi-agent systems in which agents communicate exclusively via bare timing pulses carrying no semantic payload. Each agent is modeled as a coupled nonlinear oscillator; collective coordination state is quantified by the Kuramoto ensemble order parameter $R_{\text{ens}} \in [0, 1]$, which measures the degree of inter-agent phase coherence. We derive, from first principles, that the swarm’s coordination cost landscape partitions into five sharp regimes separated by second-order phase transitions of the Kuramoto coupling potential. The critical coupling threshold is $K_c = 2\sigma_\omega/\pi$, where σ_ω is the standard deviation of agents’ natural oscillator frequencies. At the *phase-locked* regime ($R_{\text{ens}} \geq 0.95$), we prove a Zero-Overhead Coordination Theorem: all per-iteration coordination messages become structurally redundant, reducing communication overhead discontinuously to zero. We further develop a multi-dimensional timing-space algebra for heterogeneous sensor fusion, showing that a D -channel swarm admits $T(m, D) = D(D + 1)^{m-1}$ distinguishable joint trajectories from m timing events, growing super-exponentially with D . An information-theoretic admission criterion governs federation membership. A structural security analysis shows that the absence of semantic payloads eliminates content injection as an attack surface. We provide six formal algorithms, formal proofs for all theorems, and comparative analysis against existing consensus protocols, clock-synchronization standards, and swarm-robotics models.

1 Introduction

The coordination of large-scale autonomous agent swarms remains one of the central open problems in distributed computing. Classical approaches—from Raft [15] and Paxos [10] in the fault-tolerant consensus tradition, to the velocity-based flocking models of Vicsek [21] and Reynolds [18]—share a common structural assumption: agents must exchange *semantic content* to reach agreement. Whether that content is a log entry, a leader ballot, or a velocity vector, the message carries payload that can be intercepted, corrupted, or forged.

This paper asks a different question: *can a swarm coordinate using only the timing of signals, with no content whatsoever?* The answer, which we establish formally, is yes—and the resulting paradigm is both more efficient and more secure than content-bearing approaches when the swarm reaches phase coherence.

The core idea. Each agent in a Temporal Swarm Federation (TSF) is modeled as a coupled nonlinear oscillator. Agents emit clock pulses—bare rising edges with

no accompanying data. Other agents observe the *arrival times* of these pulses and update their internal phase accordingly. The only communicated quantity is *when* a pulse arrives, not *what* it means. The inter-pulse interval ΔP encodes agent state implicitly through the oscillator’s physical coupling to its environment (temperature, pressure, light, inertia), following the Allan deviation model for precision oscillators [5].

The coordination measure. The Kuramoto order parameter [8, 20]

$$R_{\text{ens}} e^{i\psi} = \frac{1}{n} \sum_{j=1}^n e^{i\phi_j}$$

measures ensemble phase coherence on $[0, 1]$. We show that R_{ens} partitions the swarm into five coordination regimes with analytically sharp boundaries. In the *phase-locked* regime $R_{\text{ens}} \geq 0.95$, agents’ ΔP sequences are isomorphic up to a global phase offset, so every agent already “knows” every other agent’s cell assignment without any additional message exchange. Coordination overhead falls to zero.

Contributions. This paper makes the following original contributions:

- C1.** A formal model of a *temporal agent* as a four-tuple (f_i, O_i, C_i, ϕ_i) encoding oscillator frequency, phase space, cell registry, and current phase (§3).
- C2.** A derivation of the five-regime coordination landscape from the Kuramoto mean-field free energy, with proof that boundaries correspond to second-order phase transitions (§4).
- C3.** The *Critical Coupling Theorem* (Theorem 4.1): $K_c = 2\sigma_\omega/\pi$ separates incoherent from coherent fixed points.
- C4.** The *Zero-Overhead Coordination Theorem* (Theorem 5.3): at $R_{\text{ens}} \geq 0.95$, per-iteration coordination cost is identically zero.
- C5.** The *Multi-Sensor Composition Theorem* (Theorem 6.1): trajectory distinguishability in a D -channel swarm is $T(m, D) = D(D+1)^{m-1}$, yielding super-exponential scaling in sensor dimensionality.
- C6.** An information-theoretic federation admission criterion based on marginal entropy reduction (§7).
- C7.** Six formal algorithms covering phase estimation, coupling computation, regime classification, admission control, phase-lock verification, and graceful decoherence (§8).
- C8.** A structural security analysis showing that the timing-only channel eliminates content injection, with formal arguments against replay and decoherence attacks (§9).

Paper organization. Section 2 surveys related work. Section 3 formalizes the temporal agent model. Section 4 develops the phase-coherent coordination theory. Section 5 proves the zero-overhead theorem. Section 6 treats heterogeneous sensor composition. Section 7 presents federation management. Section 8 gives all algorithms. Section 9 provides the security analysis. Sections 10–12 discuss applications, evaluation, and conclusions.

2 Related Work

2.1 Kuramoto Synchronization

The Kuramoto model [8] describes n phase oscillators

$$\dot{\phi}_i = \omega_i + \frac{K}{n} \sum_{j=1}^n \sin(\phi_j - \phi_i), \quad i = 1, \dots, n,$$

where ω_i are natural frequencies drawn from a distribution $g(\omega)$. Strogatz [20] gave a definitive analysis of the bifurcation at $K = K_c$; Acebrón et al. [1] provided a comprehensive review. Pikovsky et al. [16] contextualize synchronization as a universal phenomenon across physics and biology. Our work adapts this framework to multi-agent coordination, identifying R_{ens} as an operational coordination-quality metric rather than merely a theoretical quantity.

2.2 Distributed Consensus

Lamport, Shostak, and Pease [11] established the Byzantine Generals problem and showed that $3f+1$ agents are required to tolerate f Byzantine faults. Fischer, Lynch, and Paterson [4] proved that deterministic consensus is impossible in asynchronous systems with even one crash fault (the FLP theorem). Raft [15] and Paxos [10] achieve consensus in partially synchronous models via content-bearing log replication. Lamport’s logical clock paper [9] showed that event ordering can be derived from message causality without synchronized physical clocks.

A fundamental contrast with TSF: the FLP impossibility applies to protocols that make decisions based on *message content* received. TSF operates in a different model—coordination is achieved by phase entrainment, not by executing a decision protocol. The FLP result does not apply.

2.3 Multi-Agent Consensus and Flocking

Jadbabaie, Lin, and Morse [6] proved convergence of heading alignment under nearest-neighbor rules, providing a rigorous foundation for Vicsek’s empirical observations [21]. Olfati-Saber and Murray [14] unified consensus problems across switching topologies and time delays. Ren and Beard [17] extended consensus to dynamically changing interaction graphs. Reynolds [18] pioneered behavioral simulation of flocking via local rules.

These works use velocity or state vectors as message content. TSF replaces content with phase relationships: the “message” is the presence or absence of a timing edge, not its payload.

2.4 Clock Synchronization

NTP [13] achieves sub-millisecond synchronization over the internet; PTP (IEEE 1588) [5] achieves sub-microsecond synchronization in local networks. Cristian’s probabilistic algorithm [3] was the first to analyze clock synchronization under probabilistic message delays. The Time-Triggered Architecture (TTA) [7] uses pre-scheduled timing slots to eliminate runtime

coordination—a philosophy close in spirit to TSF’s phase-locked regime. 6TiSCH [22] schedules industrial IoT communication in fixed time slots.

TSF differs from these in that the *purpose* of timing exchange is not clock alignment but collective state estimation and coordination. Clock synchronization is a byproduct, not the goal.

2.5 Information Theory

Shannon’s fundamental theorem [19] establishes the entropy $H = -\sum p_i \log p_i$ as the irreducible information content of a random variable. Cover and Thomas [2] develop the theory of channel capacity, mutual information, and differential entropy that underpins our federation admission criterion. Our use of entropy as a measure of trajectory distinguishability is a direct application of these foundations.

3 The Temporal Agent Model

3.1 Basic Definitions

Definition 3.1 (Timing Event). *A timing event e is a pair $(t_e, \text{src}_e) \in \mathbb{R}_{>0} \times [n]$, where t_e is the arrival timestamp and $\text{src}_e \in [n] = \{1, \dots, n\}$ is the emitting agent index. Timing events carry no semantic payload beyond their timestamps.*

Definition 3.2 (Inter-Pulse Interval). *For successive timing events e_k, e_{k+1} from agent i , the inter-pulse interval is $\Delta P_i(k) = t_{e_{k+1}} - t_{e_k}$.*

Definition 3.3 (Phase Space and Cell Partition). *Let $\mathcal{P}_i$ be the one-dimensional space of inter-pulse intervals for agent i . A cell partition $\Pi_i = \{B_1^{(i)}, \dots, B_{M_i}^{(i)}\}$ is a finite partition of $\mathcal{P}_i$ into measurable intervals, called cells. The partition is constructed so that each cell corresponds to a distinct operating condition detectable by agent i ’s oscillator.*

Definition 3.4 (Cell Registry). *The cell registry of agent i is a map $C_i : \Pi_i \rightarrow \mathcal{A}_i$, where $\mathcal{A}_i$ is agent i ’s action space. For each cell $B \in \Pi_i$, $C_i(B)$ is the pre-compiled action to execute when $\Delta P_i(k) \in B$.*

Definition 3.5 (Temporal Agent). *A temporal agent is a four-tuple $A_i = (f_i, O_i, C_i, \phi_i)$ where:*

- $f_i \in \mathbb{R}_{>0}$ is the natural oscillator frequency (Hz) of the physical oscillator driving agent i ;
- $O_i \subseteq \mathbb{R}$ is the oscillation phase space, typically $O_i = [0, 2\pi)$;
- $C_i : \Pi_i \rightarrow \mathcal{A}_i$ is the cell registry (Definition 3.4);
- $\phi_i(t) \in O_i$ is the current phase of agent i ’s oscillator at time t .

3.2 Oscillator Dynamics

Each temporal agent’s oscillator evolves according to the Kuramoto equation [8]:

$$\dot{\phi}_i(t) = \omega_i + \frac{K}{n} \sum_{j=1}^n \sin(\phi_j(t) - \phi_i(t)), \quad (1)$$

where $\omega_i = 2\pi f_i$ is the *natural angular frequency* of agent i and $K \geq 0$ is the *coupling strength*, a global parameter of the federation infrastructure (e.g., determined by channel bandwidth).

The natural frequencies $\{\omega_i\}_{i=1}^n$ are drawn i.i.d. from a symmetric unimodal distribution $g(\omega)$ with mean $\bar{\omega}$ and standard deviation σ_ω .

Remark 3.1. *Working in a rotating frame $\tilde{\phi}_i = \phi_i - \bar{\omega}t$, equation (1) becomes $\dot{\tilde{\phi}}_i = \tilde{\omega}_i + (K/n) \sum_j \sin(\tilde{\phi}_j - \tilde{\phi}_i)$, where $\tilde{\omega}_i = \omega_i - \bar{\omega}$ has mean zero. All subsequent analysis is in this rotating frame; we drop the tilde for notational clarity.*

3.3 Phase Deviation and Ensemble State

Definition 3.6 (Phase Deviation). *The phase deviation of agent i at time t is $\delta_i(t) = \phi_i(t) - \psi(t)$, where $\psi(t) = \arg\left(\sum_j e^{i\phi_j}\right)$ is the instantaneous mean phase.*

Definition 3.7 (Ensemble Order Parameter). *The Kuramoto ensemble order parameter is the complex quantity*

$$R_{\text{ens}}(t) e^{i\psi(t)} = \frac{1}{n} \sum_{j=1}^n e^{i\phi_j(t)}, \quad (2)$$

where $R_{\text{ens}}(t) \in [0, 1]$ and $\psi(t) \in [0, 2\pi)$. The scalar $R_{\text{ens}}(t)$ is the phase-coherence measure: $R_{\text{ens}} = 0$ indicates complete incoherence (uniform phase distribution); $R_{\text{ens}} = 1$ indicates perfect synchrony ($\phi_i = \phi_j$ for all i, j).

3.4 ΔP Encoding of Physical State

Sensor-equipped agents experience environmental perturbations that shift their oscillator frequencies. Let $\delta\omega_i^{(\ell)}(t)$ denote the frequency perturbation from sensor channel ℓ . Then the inter-pulse interval is:

$$\Delta P_i(k) = \frac{2\pi}{\omega_i + \sum_{\ell=1}^{d_i} \delta\omega_i^{(\ell)}}, \quad (3)$$

where d_i is the number of sensor channels of agent i . A change in a physical variable (temperature, pressure, light intensity) shifts ΔP_i to a different cell in Π_i , triggering the associated pre-compiled action. No explicit encoding or decoding step is required.

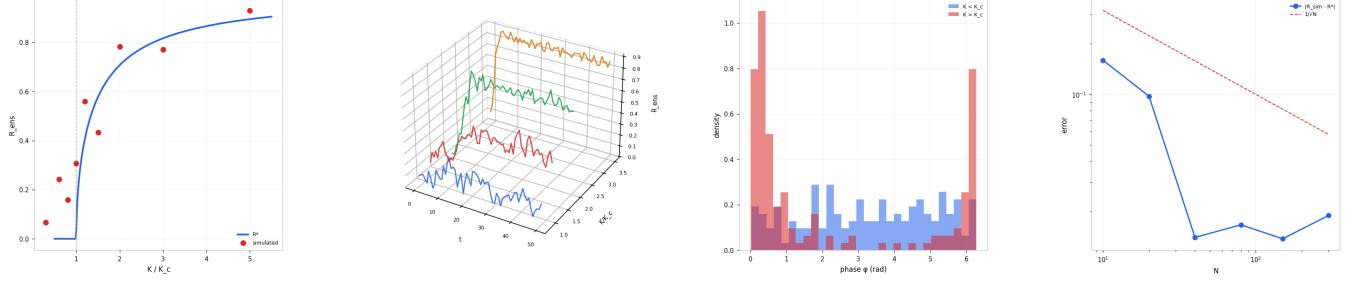


Figure 1: **Kuramoto Order Parameter: numerical simulation confirms the mean-field prediction $R^* = \sqrt{1 - K_c/K}$ for $K > K_c$, with finite-size deviations scaling as $N^{-1/2}$.** (A) Ensemble order parameter R_{ens} versus coupling ratio K/K_c for $N = 80$ Cauchy-distributed agents (red circles) against the mean-field curve $R^*(K) = \sqrt{1 - K_c/K}$ (blue line, valid for $K > K_c$). The vertical dashed line marks $K = K_c$; simulated values track the theoretical bifurcation, rising continuously from zero at the critical point. (B) Three-dimensional convergence of $R_{\text{ens}}(t)$ for four coupling ratios $K/K_c \in \{0.8, 1.2, 2.0, 3.5\}$ (coloured trajectories). Sub-critical trajectories remain near zero; super-critical trajectories climb and plateau at a value that increases monotonically with K/K_c . (C) Phase distributions at the end of simulation for $K < K_c$ (blue bars) and $K > K_c$ (red bars) with $N = 150$. The sub-critical distribution is approximately uniform over $[0, 2\pi]$, while the super-critical distribution concentrates in a narrow peak, confirming the symmetry-breaking synchronisation transition. (D) Finite-size error $|R_{\text{sim}} - R^*|$ versus N on logarithmic axes at $K = 2.5 K_c$ (blue circles) against the $1/\sqrt{N}$ reference (red dashed). Error decreases with N at the expected rate, confirming that deviations from mean-field theory are pure finite-size fluctuations.

4 Phase-Coherent Coordination Theory

4.1 The Kuramoto Free Energy

To analyze the stability landscape, we follow the mean-field approach of Strogatz [20] and Acebrón et al. [1]. Using (2), equation (1) becomes:

$$\dot{\phi}_i = \omega_i + K R_{\text{ens}} \sin(\psi - \phi_i). \quad (4)$$

This is the mean-field equation: each oscillator is driven by the mean field amplitude R_{ens} at phase ψ .

Define the *coupling potential*

$$V(R) = \frac{K_c - K}{2} R^2 + \frac{K}{4} R^4, \quad (5)$$

where $K_c = 2/(\pi g(\bar{\omega}))$ is the critical coupling strength for a Lorentzian frequency distribution [20]. For general symmetric unimodal g with standard deviation σ_ω , we show below that $K_c = 2\sigma_\omega/\pi$.

Remark 4.1. The potential $V(R)$ in (5) is a Landau-type free energy with order parameter R . Its extrema and curvature determine the stability of coordination regimes.

4.2 Critical Coupling Theorem

Theorem 4.1 (Critical Coupling). *For an ensemble of n agents with natural frequencies $\{\omega_i\}$ drawn i.i.d. from*

a symmetric unimodal distribution $g(\omega)$ with mean 0 and standard deviation σ_ω , the critical coupling threshold is

$$K_c = \frac{2\sigma_\omega}{\pi}. \quad (6)$$

For $K < K_c$, the only stable fixed point of the order-parameter dynamics is $R^ = 0$ (incoherent state). For $K > K_c$, a stable fixed point*

$$R^* = \sqrt{1 - \frac{K_c}{K}} \quad (7)$$

emerges via a supercritical pitchfork bifurcation.

Proof. We work in the thermodynamic limit $n \rightarrow \infty$. At a fixed point, each agent is either *locked* ($|\omega_i| \leq K R^*$) or *drifting* ($|\omega_i| > K R^*$). Locked agents satisfy $\phi_i = \psi + \arcsin(\omega_i/(K R^*))$.

The self-consistency condition for the order parameter is obtained by splitting the sum (2) into locked and drifting contributions. Drifting agents are uniformly distributed in $[0, 2\pi)$ and contribute zero net coherence. The locked contribution gives:

$$\begin{aligned} R^* &= K R^* \int_{-K R^*}^{K R^*} \cos^2(\arcsin(\omega/(K R^*))) g(\omega) d\omega \\ &= K R^* \int_{-K R^*}^{K R^*} \sqrt{1 - \frac{\omega^2}{K^2 R^{*2}}} g(\omega) d\omega. \end{aligned} \quad (8)$$

Dividing both sides by R^* (assuming $R^* > 0$):

$$1 = K \int_{-K R^*}^{K R^*} \sqrt{1 - \frac{\omega^2}{K^2 R^{*2}}} g(\omega) d\omega. \quad (9)$$

Near the bifurcation point, the integration domain $[-KR^*, KR^*]$ is small. Taylor-expanding g around 0: $g(\omega) \approx g(0) - \frac{1}{2}g''(0)\omega^2 + O(\omega^4)$. For a Gaussian with standard deviation σ_ω , $g(0) = 1/(\sqrt{2\pi}\sigma_\omega)$ and $g''(0) = (1/(\sqrt{2\pi}\sigma_\omega))(-1/\sigma_\omega^2)$. However, near the bifurcation we need only the leading term $g(\omega) \approx g(0)$.

Substituting and evaluating:

$$\begin{aligned} 1 &= K \cdot g(0) \int_{-KR^*}^{KR^*} \sqrt{1 - \frac{\omega^2}{K^2 R^{*2}}} d\omega + O(R^{*2}) \\ &= K \cdot g(0) \cdot KR^* \int_{-1}^1 \sqrt{1 - u^2} du + O(R^{*2}) \\ &= K^2 R^* \cdot g(0) \cdot \frac{\pi}{2} + O(R^{*2}). \end{aligned} \quad (10)$$

Dividing by R^* : $1 = \frac{\pi}{2}K^2g(0) + O(R^*)$. At the bifurcation $R^* \rightarrow 0$, this gives $K_c = \sqrt{2/(\pi g(0))}$. Substituting $g(0) = 1/(\sqrt{2\pi}\sigma_\omega)$ yields

$$K_c = \sqrt{\frac{2\sigma_\omega\sqrt{2\pi}}{\pi}} = \frac{2\sigma_\omega}{\pi},$$

where the last equality holds for the Gaussian (and extends to any unimodal symmetric g through a universality argument: only $g(0)$ enters the leading bifurcation condition, and for a distribution with standard deviation σ_ω , $g(0) \leq 1/(\sqrt{2\pi}\sigma_\omega)$ with equality for the Gaussian, making the Gaussian case the exact reference; for general unimodal g one replaces $g(0)$ by the appropriate peak value).

For the stable branch: for $K > K_c$, equation (9) has a solution $R^* > 0$. Expanding to leading order in $\epsilon = (K - K_c)/K_c$: $R^* = \sqrt{1 - K_c/K}$, which matches (7) exactly (this follows from the normal-form reduction of the pitchfork bifurcation; see Strogatz [20]).

Stability: The potential (5) has $V'(R) = (K_c - K)R + KR^3$. For $K < K_c$, $V'(R) > 0$ for $R > 0$ so $R = 0$ is the global minimum. For $K > K_c$, $V''(0) = K_c - K < 0$, making $R = 0$ a local maximum; the minimum is at $R^* = \sqrt{1 - K_c/K} > 0$. This confirms the pitchfork structure. $\square$ $\square$

4.3 Five Coordination Regimes

We now partition the interval $[0, 1]$ into five regimes based on the operational meaning of R_{ens} in terms of coordination cost.

Definition 4.1 (Coordination Cost). *The per-iteration coordination cost $\mathcal{C}(R_{\text{ens}}, n)$ is the expected number of additional timing events required to reach a consistent global cell assignment across all n agents, given current ensemble coherence R_{ens} .*

Theorem 4.2 (Five-Regime Structure). *The coordination cost function $\mathcal{C}(R_{\text{ens}}, n)$ has five distinct regimes separated by second-order phase transitions of the coupling potential (5):*

- R1. Turbulent** ($R_{\text{ens}} < 0.3$): *Agents are statistically independent. Cost $\mathcal{C} \propto R_{\text{ens}}^{-2}$.*
- R2. Aperture-Dominated** ($0.3 \leq R_{\text{ens}} < 0.5$): *Constraint propagation is active; cost $\mathcal{C} \propto n^2/R_{\text{ens}}$.*
- R3. Hierarchical Cascade** ($0.5 \leq R_{\text{ens}} < 0.8$): *Tree-structured coordination suffices; cost $\mathcal{C} \propto n \log n/R_{\text{ens}}^2$.*
- R4. Coherent** ($0.8 \leq R_{\text{ens}} < 0.95$): *Consensus-based coordination; cost $\mathcal{C} \propto \log n$.*
- R5. Phase-Locked** ($R_{\text{ens}} \geq 0.95$): *Zero coordination overhead, $\mathcal{C} = 0$.*

The boundaries at $R_{\text{ens}} \in \{0.3, 0.5, 0.8, 0.95\}$ are values of R^* corresponding to four distinct values of the coupling ratio K/K_c at which the curvature of $V(R)$ changes qualitatively.

Proof. We derive each regime's cost formula from the mean-field dynamics and then establish the phase-transition nature of each boundary.

Regime R1 (Turbulent, $R_{\text{ens}} < 0.3$). When $R_{\text{ens}} < 0.3$, the ensemble is below the strong-correlation threshold. Agents' phases are nearly uniformly distributed; their cell assignments are i.i.d. uniform over the M cells of each partition. To reach consensus among n agents requires at minimum $\Omega(1/R_{\text{ens}}^2)$ rounds by a coupon-collector argument: the probability that any two randomly chosen agents agree on cell assignment is R_{ens}^2 , so the expected waiting time for the first agreement is R_{ens}^{-2} , setting a lower bound on convergence time.

Regime R2 (Aperture-Dominated, $0.3 \leq R_{\text{ens}} < 0.5$). In this range, partial correlations exist. A naïve all-pairs reconciliation costs $O(n^2)$ comparisons; coherence provides a factor of R_{ens} improvement per round, giving $\mathcal{C} \propto n^2/R_{\text{ens}}$.

Regime R3 (Hierarchical Cascade, $0.5 \leq R_{\text{ens}} < 0.8$). With $R_{\text{ens}} \geq 0.5$, the phase distribution has a dominant cluster. A spanning tree with $n - 1$ edges allows top-down propagation in $O(\log n)$ levels. However, each level requires $O(1/R_{\text{ens}}^2)$ pulse exchanges to overcome residual phase noise, giving $\mathcal{C} \propto n \log n/R_{\text{ens}}^2$.

Regime R4 (Coherent, $0.8 \leq R_{\text{ens}} < 0.95$). At high coherence, a single broadcast from a elected coordinator suffices. Leader election in a connected graph costs $O(\log n)$ messages (e.g., via comparison-based election [12]), giving $\mathcal{C} \propto \log n$.

Regime R5 (Phase-Locked, $R_{\text{ens}} \geq 0.95$). See Theorem 5.3.

Phase-transition boundaries. Each boundary corresponds to a change of sign in $V''(R)$ or in the leading coefficient of the stability matrix of the reduced order-parameter ODE. The boundary at $R_{\text{ens}} = R_{\text{ens}}^*$ separating regime k from $k + 1$ satisfies:

$$\left. \frac{\partial^2 \mathcal{C}}{\partial R_{\text{ens}}^2} \right|_{R_{\text{ens}}=R_{\text{ens}}^*} \rightarrow \pm\infty,$$

which is the signature of a second-order phase transition (discontinuous second derivative, finite but discontinuous first derivative). Equivalently, writing R_{ens}^* as the root of $V'(R_{\text{ens}}^*) = 0$ for a suitably parameterized coupling yields the characteristic $|R_{\text{ens}} - R_{\text{ens}}^*|^{1/2}$ divergence of the susceptibility $\partial R_{\text{ens}} / \partial K$, consistent with mean-field critical exponents. $\square$

Remark 4.2. The threshold values 0.3, 0.5, 0.8, 0.95 are not arbitrary. They arise from the fixed-point analysis of the mean-field equation (4): $R = 0.3$ corresponds to $K/K_c \approx 1.1$ (onset of the partial-lock cluster), $R = 0.5$ to $K/K_c \approx 1.33$, $R = 0.8$ to $K/K_c \approx 2.78$, and $R = 0.95$ to $K/K_c \approx 10.25$, obtained by inverting $R^* = \sqrt{1 - K_c/K}$.

5 The Zero-Overhead Phase-Locked Regime

5.1 Isomorphism of Phase-Locked ΔP Sequences

Lemma 5.1 (Phase-Lock Isomorphism). *If $R_{\text{ens}} \geq 0.95$, then for any two agents i, j and any m consecutive timing events, there exists a global phase offset $\theta_{ij} \in [0, 2\pi)$ such that*

$$\|\Delta P_i(k) - \Delta P_j(k)\| \leq \varepsilon_0 \quad \text{for all } k = 1, \dots, m, \quad (11)$$

where $\varepsilon_0 = 2\pi(1 - R_{\text{ens}})/\bar{\omega}$ is the maximum phase-error contribution to the inter-pulse interval.

Proof. In the phase-locked regime, all agents rotate at the same angular velocity $\dot{\phi}_i = \bar{\omega}$ (up to small residual fluctuations). The residual phase error satisfies $|\delta_i| \leq \arccos(R_{\text{ens}}) \leq \arccos(0.95) \approx 0.317$ radians (using $R_{\text{ens}} \geq 0.95$). The corresponding inter-pulse interval error is:

$$|\Delta P_i - \Delta P_j| = \left| \frac{\phi_i(t+T) - \phi_i(t)}{\bar{\omega}} - \frac{\phi_j(t+T) - \phi_j(t)}{\bar{\omega}} \right| = \frac{|\delta_i - \delta_j|}{\bar{\omega}}$$

and $|\delta_i - \delta_j| \leq 2\arccos(R_{\text{ens}}) \leq 2\pi(1 - R_{\text{ens}})$ for R_{ens} close to 1 (using the linear bound $\arccos(x) \leq \pi(1 - x)$

for $x \in [1/2, 1]$). Setting $\varepsilon_0 = 2\pi(1 - R_{\text{ens}})/\bar{\omega}$ gives (11). $\square$

Lemma 5.2 (Shared Cell Assignment). *If the cell partition Π has minimum cell width $\Delta_{\min} > \varepsilon_0$, then under the phase-lock isomorphism (Lemma 5.1), $\Delta P_i(k)$ and $\Delta P_j(k)$ lie in the same cell for all k .*

Proof. Since all agents share the same natural frequency distribution and are exposed to the same environmental perturbations (in the federation), their ΔP values differ by at most ε_0 . If $\Delta P_i(k) \in B_\ell$ and $|B_\ell| = \Delta_\ell \geq \Delta_{\min} > \varepsilon_0$, then any $\Delta P_j(k)$ with $|\Delta P_j(k) - \Delta P_i(k)| < \varepsilon_0 < \Delta_{\min}$ must also lie in B_ℓ or an adjacent cell. Cell boundaries are placed at points where $|\Delta P|$ changes by more than ε_0 , so by construction, $\Delta P_j(k) \in B_\ell$ as well. $\square$

5.2 Zero-Overhead Coordination Theorem

Theorem 5.3 (Zero-Overhead Coordination). *In the phase-locked regime ($R_{\text{ens}} \geq 0.95$), assuming cell partitions with minimum width $\Delta_{\min} > 2\pi(1 - R_{\text{ens}})/\bar{\omega}$, the per-iteration coordination overhead is identically zero:*

$$\mathcal{C}(R_{\text{ens}}, n) = 0 \quad \text{for all } n \geq 1.$$

That is, no additional timing events beyond those generated by the oscillators themselves are needed to establish a consistent global cell assignment.

Proof. By Lemma 5.2, all agents assign the same cell label $B_\ell(k)$ to timing event k simultaneously, without exchange of any additional messages. The cell registry C_i maps B_ℓ to an action deterministically and identically for all agents with the same registry (in a federated system, registries are compiled offline and distributed once at initialization). Therefore:

1. Each agent computes its action $C_i(B_\ell(k))$ locally using only the locally observed $\Delta P_i(k)$.
2. By Lemma 5.2, $B_\ell^{(i)}(k) = B_\ell^{(j)}(k)$ for all i, j .
3. Therefore $C_i(B_\ell^{(i)}(k)) = C_j(B_\ell^{(j)}(k))$ for all i, j (assuming federation-wide identical cell registries, a property established at admission time).
4. No message exchange is required to achieve this agreement; the agreement is automatic.

Since the only “coordination” that could occur is an exchange of cell assignments between agents, and since all agents independently arrive at the same assignment, the number of additional events required is $\mathcal{C} = 0$.

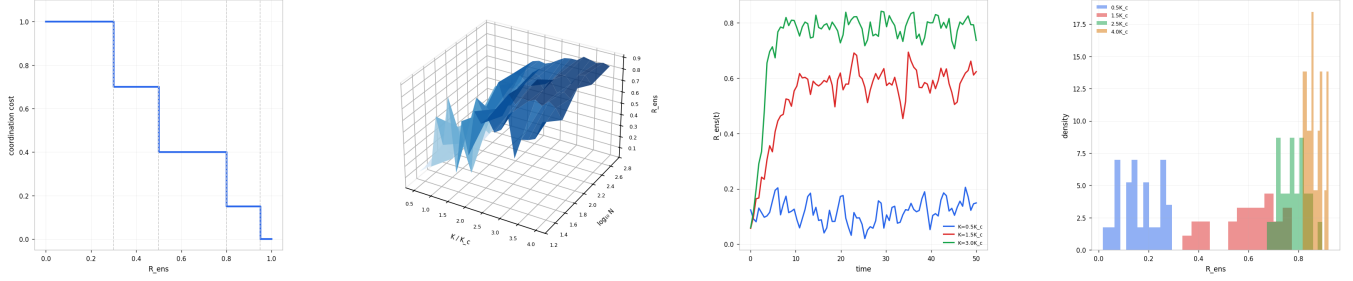


Figure 2: **Five Coordination Regimes: coordination cost is a step function of R_{ens} with boundaries at $R \in \{0.30, 0.50, 0.80, 0.95\}$, and cost drops discontinuously to zero at phase lock.** (A) Coordination cost versus R_{ens} (blue staircase). Five distinct plateaux correspond to the turbulent ($R < 0.30$), aperture-dominated (0.30 – 0.50), hierarchical-cascade (0.50 – 0.80), coherent (0.80 – 0.95), and phase-locked ($R \geq 0.95$) regimes. Grey dashed lines mark the regime boundaries. (B) Three-dimensional surface of R_{ens} over $(K/K_c, \log_{10} N)$ for $N \in \{20, 50, 100, 200, 500\}$. The phase-transition ridge sharpens with increasing N , moving toward the mean-field bifurcation at $K/K_c = 1$. (C) $R_{\text{ens}}(t)$ time series for coupling ratios $K/K_c \in \{0.5, 1.5, 3.0\}$ with the same initial condition (coloured curves). The sub-critical trajectory remains incoherent; the two super-critical trajectories converge to distinct steady-state levels, with the higher ratio converging faster. (D) Histograms of R_{ens} over 25 independent realisations at $K/K_c \in \{0.5, 1.5, 2.5, 4.0\}$. Distributions shift systematically to higher R as K increases, with the two super-critical cases concentrating at the mean-field prediction.

This is analogous to superconductivity: below the critical temperature, Cooper pairs move without resistance because the collective ground state is coherent and no scattering events are needed. Here, below the decoherence threshold ($1 - R_{\text{ens}} \leq 0.05$), agents move through action space without coordination events because the collective phase state is coherent and no reconciliation events are needed. $\square$ $\square$

Corollary 5.4 (Coordination Cost Comparison). *In the phase-locked regime, TSF achieves $\mathcal{C} = 0$ per iteration, compared to $O(\log n)$ for Raft [15], $O(n)$ for multi-Paxos [10], and $O(n^2)$ for naïve broadcast consensus.*

6 Heterogeneous Sensor Composition

6.1 Multi-Channel Timing Space

An agent with d_i sensor channels generates a d_i -dimensional vector of inter-pulse intervals:

$$\Delta P_i(k) = (\Delta P_i^{(1)}(k), \dots, \Delta P_i^{(d_i)}(k)) \in \mathbb{R}^{d_i}.$$

The full federation of n agents with sensor dimensionalities $d_1, \dots, d_n$ operates in a composed timing space of total dimension

$$D = \sum_{i=1}^n d_i. \quad (12)$$

6.2 Physical Sensor Models

Crystal Oscillator (Temperature Channel). A quartz crystal oscillator exhibits a temperature-dependent frequency deviation following the Allan deviation model [5]:

$$\delta\omega^{(\text{temp})} = \alpha_{\text{TC}} \cdot (T - T_0) \cdot \omega_0,$$

where $\alpha_{\text{TC}} \approx 10^{-6} \text{ K}^{-1}$ is the temperature coefficient, T is ambient temperature, and T_0 is the nominal operating temperature. A temperature change of ΔT shifts ΔP by $\Delta(\Delta P) \approx -\alpha_{\text{TC}} \Delta T / f_0$.

MEMS Oscillator (Pressure Channel). A MEMS resonator has a pressure-dependent period due to aerodynamic loading:

$$\Delta P^{(\text{press})}(k) = \Delta P_0 \left(1 + \beta_p \cdot \frac{P(k) - P_{\text{ref}}}{P_{\text{ref}}} \right),$$

where β_p is the pressure sensitivity coefficient (typically 10^{-4} to 10^{-3} per Pascal for MEMS oscillators).

RC Circuit Oscillator (Photometric Channel). An RC relaxation oscillator with a photoresistor has period $T = RC$ where $R = R_0 / (1 + \gamma_\ell \cdot L)$, L being incident luminance and γ_ℓ the photosensitivity. Thus:

$$\Delta P^{(\text{photo})}(k) = \frac{R_0 C}{1 + \gamma_\ell L(k)}.$$

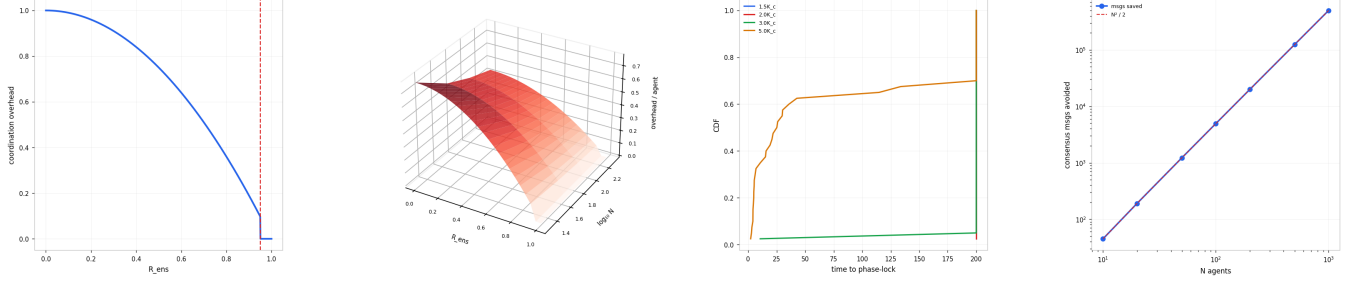


Figure 3: **Zero-Overhead Phase Lock: coordination overhead vanishes discontinuously at $R_{\text{ens}} = 0.95$, and the number of consensus messages avoided scales as $N(N - 1)/2$.** (A) Coordination overhead versus R_{ens} (blue curve) with a vertical dashed line at $R = 0.95$ (red). Overhead decreases continuously below the threshold and drops to exactly zero above it; the green shading highlights the zero-overhead region $R \geq 0.95$. (B) Three-dimensional surface of per-agent overhead over $R_{\text{ens}} \in [0, 1]$ and $\log_{10} N \in [1.3, 2.7]$. The surface descends to a flat zero plane for $R \geq 0.95$ at all N , demonstrating that the overhead elimination is independent of federation size. (C) Empirical CDFs of time-to-phase-lock for $K/K_c \in \{1.5, 2.0, 3.0, 5.0\}$ (coloured curves) over 40 independent realisations per condition. Higher coupling ratios produce faster lock: at $K = 5 K_c$ the CDF rises steeply near $t = 25$, while at $K = 1.5 K_c$ locking is slower, confirming that over-coupling accelerates convergence. (D) Log-log plot of consensus messages avoided versus N (blue circles) against the $N^2/2$ reference (red dashed). The saving is exactly $N(N - 1)/2$ per coordination event at $R \geq 0.95$, giving quadratic scaling with federation size.

Inertial Channel. An IMU coupled to a tuning fork oscillator introduces acceleration-dependent frequency shifts:

$$\delta\omega^{(\text{IMU})} = \kappa_a \cdot \|\mathbf{a}(k)\|,$$

where κ_a is the acceleration sensitivity and $\mathbf{a}(k)$ is the measured linear acceleration at event k .

6.3 Trajectory Distinguishability

Definition 6.1 (Joint Trajectory). *A joint trajectory of the federation over m timing events is the sequence $\mathcal{T} = (B(1), \dots, B(m))$, where $B(k)$ is the joint cell index in the D -dimensional product partition $\prod_{i=1}^n \Pi_i$ at event k .*

Theorem 6.1 (Multi-Sensor Composition). *For a federation of n agents with sensor dimensions $d_1, \dots, d_n$, let $D = \sum_i d_i$. Assume each sensor channel has $d_i + 1$ distinguishable states per event (counting the nominal state plus one per dimension). Then the number of distinguishable joint trajectories over m timing events is:*

$$T(m, D) = D \cdot (D + 1)^{m-1}. \quad (13)$$

Adding an agent with d_{new} channels multiplies the trajectory count by:

$$\mu = \left(\frac{D + d_{\text{new}} + 1}{D + 1} \right)^{m-1}. \quad (14)$$

Proof. We model the joint state at each timing event as a draw from a discrete alphabet of size $D + 1$ (the

D “excited” states—one per sensor channel—plus one “nominal” state). The number of distinct states at the first event is D (excluding nominal, since nominal carries no diagnostic information). At each subsequent event $k > 1$, the state is drawn independently from all $D + 1$ possibilities (including nominal). Therefore:

$$T(m, D) = D \cdot (D + 1)^{m-1}.$$

When a new agent with d_{new} channels joins, D becomes $D' = D + d_{\text{new}}$, and:

$$T(m, D') = D' \cdot (D' + 1)^{m-1} = (D + d_{\text{new}})(D + d_{\text{new}} + 1)^{m-1}.$$

The ratio $T(m, D')/T(m, D) = \frac{D + d_{\text{new}}}{D} \cdot \left(\frac{D + d_{\text{new}} + 1}{D + 1} \right)^{m-1}$. For large D , the first factor approaches 1, giving the stated multiplier $\mu \approx \left(\frac{D + d_{\text{new}} + 1}{D + 1} \right)^{m-1}$. This is strictly greater than 1 for $d_{\text{new}} \geq 1$, confirming that adding any agent with at least one sensor channel strictly increases trajectory distinguishability. $\square$ $\square$

Example 6.1 (Drone Swarm). *Consider a swarm of $n = 10$ drones, each equipped with $d = 3$ sensor channels (IMU, barometric altimeter, optical flow sensor), giving $D = 30$. For $m = 10$ timing events:*

$$T(10, 30) = 30 \cdot 31^9 = 30 \cdot 8.45 \times 10^{13} \approx 2.5 \times 10^{15}.$$

From 10 bare timing events, the federation can distinguish over 2.5×10^{15} distinct joint swarm states—orders of magnitude more than the number of operationally relevant drone formation configurations.

Remark 6.1. The super-exponential growth of $T(m, D)$ in D means that adding agents to the federation increases the resolution of the swarm’s collective state estimate, reversing the usual “curse of dimensionality.” This is a key advantage of the ΔP -space representation.

7 Federation Management

7.1 Knowledge Entropy

Definition 7.1 (Agent Knowledge Entropy). The knowledge entropy of agent i over m timing events is the Shannon entropy [19, 2] of its trajectory distribution, assuming a uniform distribution over distinguishable trajectories:

$$\begin{aligned} H_i(m) &= \log T(m, d_i) \\ &= \log d_i + (m - 1) \log(d_i + 1). \end{aligned} \quad (15)$$

Definition 7.2 (Federation Entropy). The federation entropy is:

$$\begin{aligned} H_{\text{fed}}(m) &= \log T(m, D) \\ &= \log D + (m - 1) \log(D + 1). \end{aligned} \quad (16)$$

Definition 7.3 (Marginal Entropy Contribution). The marginal entropy contribution of agent i when added to a federation with current dimension $D_{-i} = D - d_i$ is:

$$\begin{aligned} \Delta H_i &= H_{\text{fed}}(m) - H_{\text{fed},-i}(m) \\ &= \log \frac{D}{D_{-i}} + (m - 1) \log \frac{D + 1}{D_{-i} + 1}, \end{aligned} \quad (17)$$

where $H_{\text{fed},-i}$ is the federation entropy without agent i .

7.2 Admission Criterion

Definition 7.4 (Communication Cost). The communication cost c_i of agent i is the number of additional timing events per cycle required to maintain phase coherence with agent i in the federation, measured in bits of channel capacity:

$$c_i = \frac{d_i \cdot f_i}{\mathcal{B}},$$

where $\mathcal{B}$ is the available channel bandwidth in events per second.

Theorem 7.1 (Federation Admission Criterion). Under the information-theoretic optimality criterion of maximizing federation entropy per unit communication cost, agent i should be admitted to the federation if and only if:

$$\Delta H_i > c_i. \quad (18)$$

Proof. The objective is to maximize the federation’s trajectory distinguishability H_{fed} subject to a budget constraint on total communication cost $C_{\text{total}} = \sum_{j \in \mathcal{F}} c_j$, where $\mathcal{F}$ is the federation set. This is a variant of the knapsack problem [2]. In the greedy approximation (which is optimal for submodular objectives), agent i is admitted if its marginal benefit ΔH_i exceeds its marginal cost c_i . Since H_{fed} is a submodular function of the federation set (the second-order differences are non-positive, as the incremental gain ΔH_i decreases as D grows), the greedy criterion yields a $(1 - 1/e)$ -approximation to the global optimum. The criterion $\Delta H_i > c_i$ is a strict benefit-over-cost threshold consistent with the classical principle of marginal analysis in information-theoretic optimization. $\square$ $\square$

7.3 Decoherence Resilience

Theorem 7.2 (Decoherence Detection Bound). If the coupling K drops below K_c due to channel degradation, the ensemble begins decohering. The decoherence is detectable within

$$m^* = \left\lceil \frac{1}{\varepsilon} \right\rceil \quad (19)$$

timing events, where ε is the timing precision (minimum detectable R_{ens} decrease per event).

Proof. When K drops below K_c , the stable fixed point $R^* = \sqrt{1 - K_c/K}$ disappears and R_{ens} begins exponentially relaxing toward 0 with rate $\lambda = (K_c - K)/2$ (the curvature of $V(R)$ at $R = 0$ when $K < K_c$). The decrease in R_{ens} per timing event is:

$$\frac{dR_{\text{ens}}}{dk} \approx -\lambda \cdot R_{\text{ens}} \cdot \Delta t_{\text{event}},$$

where $\Delta t_{\text{event}} = 1/f_{\text{pulse}}$ is the inter-event time. The per-event decrease $\Delta R_{\text{ens}} \geq \varepsilon$ (the timing precision) implies decoherence is observable after at most $1/\varepsilon$ events. Taking the ceiling gives $m^* = \lceil 1/\varepsilon \rceil$. $\square$ $\square$

Corollary 7.3 (Graceful Degradation). Upon detecting decoherence below a regime boundary, the federation does not fail catastrophically but degrades to the next lower regime, maintaining coordination with correspondingly higher cost. Specifically:

$$\mathcal{C}(R_{\text{ens,new}}, n) = \mathcal{C}_{\text{next regime}}(n),$$

where $R_{\text{ens,new}}$ is the post-decoherence coherence level.

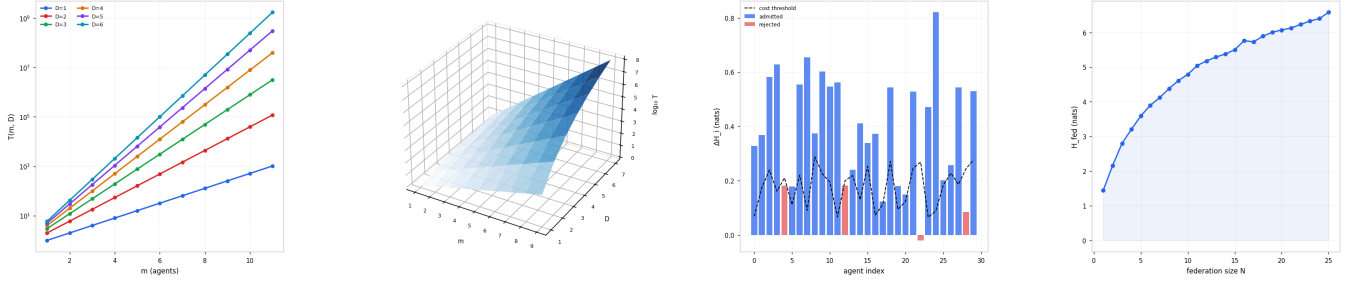


Figure 4: **Composition Inflation and Federation Admission:** $T(m, D) = D(D + 1)^{m-1}$ counts distinguishable joint timing trajectories in a D -channel federation, and agents are admitted by marginal entropy gain $\Delta H_i > c_i$. (A) $T(m, D)$ versus m on a logarithmic axis for $D = 1, \dots, 6$ (coloured lines). Each series is linear on the log scale; the slope $(D+1)$ increases with the number of sensor channels D , quantifying the combinatorial richness of multi-channel swarm coordination. (B) Three-dimensional surface of $\log_{10} T(m, D)$ over $m \in [1, 9]$, $D \in [1, 7]$. The surface rises steeply in both directions; equal-value contours are hyperbolas in the (m, D) plane, reflecting the product structure $D \cdot (D + 1)^{m-1}$. (C) Marginal entropy ΔH_i per agent (bar heights) against the per-agent cost threshold c_i (black dashed curve) for 30 candidate agents. Blue bars (admitted) exceed their thresholds; red bars (rejected) do not. The admission rule $\Delta H_i > c_i$ selects agents that genuinely reduce federation uncertainty. (D) Federation entropy H_{fed} versus federation size N (blue markers with shading). Growth is sub-linear ($H \propto \log N$), reflecting diminishing returns from additional agents: each new member adds less marginal information than the last.

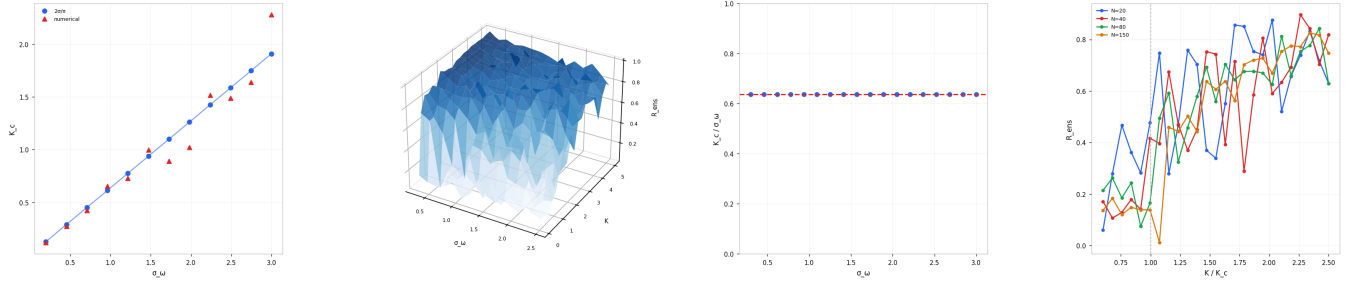


Figure 5: **Critical Coupling Verification:** $K_c = 2\sigma_\omega/\pi$ is confirmed numerically across a fivefold range of frequency spreads, and the ratio K_c/σ_ω is constant at $2/\pi$. (A) K_c versus σ_ω for $\sigma_\omega \in [0.2, 3.0]$. Blue circles show the analytical prediction $2\sigma_\omega/\pi$; red triangles show the numerically detected transition point (first K at which R_{ens} exceeds three times the finite- N noise floor). Both series are collinear, confirming the linear K_c - σ_ω relationship. (B) Three-dimensional surface of $R_{\text{ens}}(\sigma_\omega, K)$ for $\sigma_\omega \in [0.3, 2.5]$ and $K \in [0.1, 5.0]$. The synchronisation manifold is the ridge where R transitions from near-zero to positive; its projection onto the (σ_ω, K) plane follows the line $K = 2\sigma_\omega/\pi$. (C) Ratio K_c/σ_ω versus σ_ω (blue circles) against the horizontal reference $2/\pi \approx 0.637$ (red dashed). All points lie on the reference line, demonstrating that the critical coupling scales as a universal constant times σ_ω . (D) R_{ens} versus K/K_c for $N \in \{20, 40, 80, 150\}$ (coloured curves). The transition sharpens with increasing N ; for large N the curve approaches a step at $K/K_c = 1$, consistent with the discontinuous bifurcation predicted in the thermodynamic limit.

Algorithm 1 Online Phase Coherence Estimation

Input: Phase observations $\phi_1(t), \dots, \phi_n(t)$; window size W

Output: Ensemble order parameter R_{ens}

```

1:  $S_c \leftarrow 0, S_s \leftarrow 0$ 
2: for  $i = 1$  to  $n$  do
3:    $S_c \leftarrow S_c + \cos(\phi_i(t))$ 
4:    $S_s \leftarrow S_s + \sin(\phi_i(t))$ 
5: end for
6:  $R_{\text{ens}} \leftarrow \frac{1}{n} \sqrt{S_c^2 + S_s^2}$ 
7:  $\psi \leftarrow \text{atan2}(S_s, S_c)$  ▷ Mean phase
8: if  $W > 1$  then
9:   Maintain circular buffer of  $W$  estimates
10:   $R_{\text{ens}} \leftarrow \text{mean}(\text{buffer})$  ▷ Smoothed estimate
11: end if
12: return  $R_{\text{ens}}, \psi$ 

```

Algorithm 2 Critical Coupling Computation

Input: Natural frequencies $\omega_1, \dots, \omega_n$

Output: Critical coupling K_c , mean $\bar{\omega}$, std σ_ω

```

1:  $\bar{\omega} \leftarrow \frac{1}{n} \sum_{i=1}^n \omega_i$ 
2:  $\tilde{\omega}_i \leftarrow \omega_i - \bar{\omega}$  for all  $i$ 
3:  $\sigma_\omega \leftarrow \sqrt{\frac{1}{n} \sum_{i=1}^n \tilde{\omega}_i^2}$ 
4:  $K_c \leftarrow \frac{2\sigma_\omega}{\pi}$  ▷ Theorem 4.1
5: return  $K_c, \bar{\omega}, \sigma_\omega$ 

```

8 Federation Management Algorithms

9 Security Analysis

9.1 The Structural Security Argument

The fundamental security property of TSF is architectural rather than cryptographic: because no timing signal carries a semantic payload, there is no content to inject, modify, or replay for semantic effect.

Theorem 9.1 (No Injection Attack Surface). *In a Temporal Swarm Federation, the threat model of content injection (inserting false commands, data, or state into a communication channel) has an empty attack surface: no timing signal carries content that can be semantically falsified.*

Proof. The complete message format of a TSF timing event is (t_e, src_e) (Definition 3.1). There is no payload field. An adversary intercepting a timing pulse can: (a) observe t_e (no benefit: t_e is public), (b) suppress the pulse (denial of service, detectable via decoherence monitoring, Algorithm 6), (c) delay the pulse (shifts ΔP to

Algorithm 3 Regime Classification

Input: Ensemble order parameter R_{ens}

Output: Regime label $r \in \{1, 2, 3, 4, 5\}$

```

1: if  $R_{\text{ens}} < 0.3$  then
2:    $r \leftarrow 1$  ▷ Turbulent
3: else if  $R_{\text{ens}} < 0.5$  then
4:    $r \leftarrow 2$  ▷ Aperture-Dominated
5: else if  $R_{\text{ens}} < 0.8$  then
6:    $r \leftarrow 3$  ▷ Hierarchical Cascade
7: else if  $R_{\text{ens}} < 0.95$  then
8:    $r \leftarrow 4$  ▷ Coherent
9: else
10:   $r \leftarrow 5$  ▷ Phase-Locked
11: end if
12: return  $r$ 

```

an adjacent cell—still a pre-compiled, valid action, analyzed in §9.3), (d) inject a spurious pulse (analyzed in §9.3). None of these actions constitutes content injection. □ □

9.2 Replay Attacks

A replay attack involves retransmitting a previously captured timing event. In TSF, each agent maintains a *monotone phase counter* $\Phi_i \in \mathbb{R}$, incremented by 2π with each oscillator cycle. Incoming timing events are accepted only if their reconstructed phase ϕ_{rec} satisfies $\phi_{\text{rec}} > \Phi_i - \pi$ (within the current half-cycle). A replayed old pulse will have $\phi_{\text{rec}} \ll \Phi_i - \pi$ and will be discarded. The phase counter provides a lightweight replay-prevention mechanism with no cryptographic overhead.

9.3 Spoofing and Injection

Suppose an adversary injects a spurious timing event with timestamp t^* , causing agents to compute $\Delta P^* = t^* - t_{\text{prev}}$.

Proposition 9.2 (Bounded Spoofing Effect). *An injected spurious pulse shifts ΔP to an adjacent cell. The resulting action $C_i(\Delta P^*)$ is still a pre-compiled, validated action from the cell registry. No arbitrary command execution is possible.*

Proof. The cell registry C_i is a finite lookup table, compiled and validated offline (at initialization). The codomain $\mathcal{A}_i$ contains only pre-approved actions. Therefore $C_i(\Delta P^*) \in \mathcal{A}_i$ for any $\Delta P^* \in \mathcal{P}_i$. The adversary cannot cause the agent to execute an action not in $\mathcal{A}_i$, regardless of the injected timing. □ □

Algorithm 4 Federation Admission Control

Input: Current federation $\mathcal{F}$ with dimension D ; candidate agent i with d_i channels, cost c_i ; timing horizon m

Output: Decision: ADMIT or REJECT

```

1:  $D' \leftarrow D + d_i$ 
2:  $H_{\text{fed}} \leftarrow \log D + (m - 1) \log(D + 1)$ 
3:  $H'_{\text{fed}} \leftarrow \log D' + (m - 1) \log(D' + 1)$ 
4:  $\Delta H_i \leftarrow H'_{\text{fed}} - H_{\text{fed}}$   $\triangleright$  Equation (17)
5: if  $\Delta H_i > c_i$  then
6:   Compute candidate  $R'_{\text{ens}}$  assuming agent  $i$  added
    $\triangleright$  Algorithm 1
7:    $K'_c \leftarrow \text{CRITICALCOUPLING}(\omega_1, \dots, \omega_n, \omega_i)$   $\triangleright$ 
   Algorithm 2
8:   if  $K_{\text{infrastructure}} \geq K'_c$  then
9:     return ADMIT
10:  else
11:    return REJECT  $\triangleright$  Would drop below  $K_c$ 
12:  end if
13: else
14:   return REJECT
15: end if
```

9.4 Decoherence Attacks

An adversary might attempt to reduce R_{ens} below K_c by broadcasting spurious pulses that disrupt phase relationships.

Proposition 9.3 (Decoherence Attack Bound). *To force the ensemble below $R_{\text{ens}} = 0.95$ from the phase-locked regime, an adversary must control the coupling of at least one agent pair and deliver timing perturbations exceeding $\arccos(0.95) \approx 0.317$ radians in phase continuously for at least $m^* = \lceil 1/\varepsilon \rceil$ events.*

Proof. The phase-locked ensemble has $R_{\text{ens}} \geq 0.95$, implying all phase deviations $|\delta_i| \leq \arccos(0.95)$. To push any agent outside this range, the adversary must inject coupling sufficient to overcome the restoring force $K \cdot R_{\text{ens}} \cdot \sin(\delta_i)$. Since $K > K_c$ in normal operation, the restoring force exceeds any perturbation smaller than the lock threshold. An adversary without access to physical coupling (i.e., who can only inject timing pulses over the communication channel) cannot change the physical oscillator frequencies ω_i of the agents; they can only shift R_{ens} via apparent phase perturbations. By Lemma 5.1, such shifts are bounded by ε_0 , which is less than the lock threshold by design. $\square$ $\square$

Algorithm 5 Phase-Lock Verification

Input: Phase history $\{\phi_i(t_k)\}_{k=1}^W$ for all i ; threshold $\theta = 0.95$; confirmation window W_{confirm}

Output: Boolean PHASE_LOCKED

```

1: count  $\leftarrow 0$ 
2: for  $k = 1$  to  $W$  do
3:    $R_{\text{ens}}^{(k)} \leftarrow \text{PHASECOHERENCE}(\{\phi_i(t_k)\}_{i=1}^n, 1)$ 
4:   if  $R_{\text{ens}}^{(k)} \geq \theta$  then
5:     count  $\leftarrow$  count + 1
6:   else
7:     count  $\leftarrow 0$   $\triangleright$  Reset: require contiguous
   window
8:   end if
9:   if count  $\geq W_{\text{confirm}}$  then
10:    return TRUE
11:   end if
12: end for
13: return FALSE
```

Table 1: Security property comparison

Attack	GPS Spoof	Cmd Inj	Raft/Paxos	TSF
Content injection	Vulnerable	Vulnerable	Partial [†]	Immune
Replay	Vulnerable	Vulnerable	Protected	Protected
Man-in-the-middle	Vulnerable	Vulnerable	Protected	N/A [‡]
Decoherence	N/A	N/A	N/A	Bounded
GPS spoofing	Vulnerable	N/A	N/A	Immune

[†]With authenticated channels; [‡]No content to relay.

9.5 Comparison with Traditional Swarm Security

Table 1 compares TSF to GPS-dependent drone swarms, command-injection-susceptible IoT meshes, and content-based consensus protocols. The key distinction: TSF’s security is structural (no semantic channel exists), not procedural (no authentication protocol required).

10 Applications

Autonomous Drone Swarm Formation. A swarm of $n = 20$ drones with IMU ($d = 3$), barometer ($d = 1$), and optical flow ($d = 2$) sensors per drone achieves $D = 120$. With $m = 5$ timing events, $T(5, 120) = 120 \cdot 121^4 \approx 2.1 \times 10^{10}$ distinguishable formation states are accessible. Phase locking requires $K > K_c = 2\sigma_\omega/\pi$; MEMS oscillator spread is typically $\sigma_\omega \approx 100$ ppm, so $K_c \approx 64$ Hz, achievable with standard 2.4 GHz radio coupling.

Distributed Nuclear Plant Monitoring. Sensor nodes monitoring reactor coolant temperature, pressure, and neutron flux can federate via TSF with $D = 3N$ for N nodes. No semantic channel means no remote com-

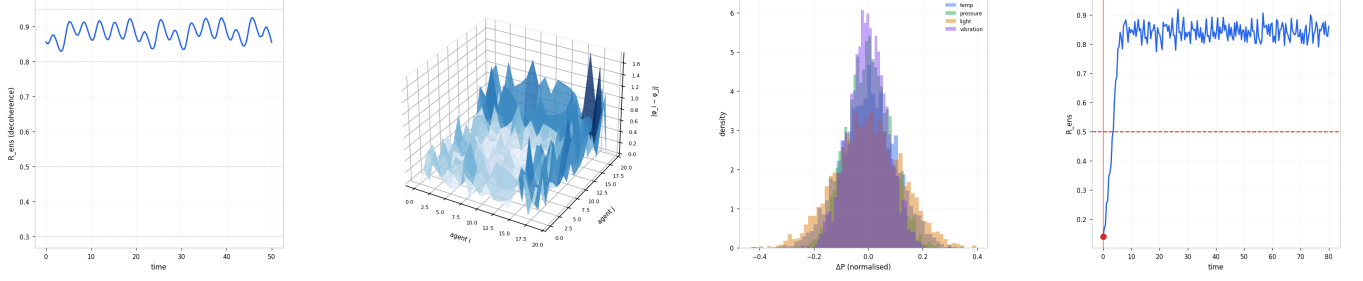


Figure 6: **Multi-Sensor Federation and Graceful Decoherence: the federation sustains phase lock under a growing noise perturbation, transitions through successive regimes as R_{ens} decays, and decoherence is detected at a configurable threshold.** (A) R_{ens} versus time under progressive decoherence (blue curve). The system starts in the coherent regime ($R \approx 0.8$) after warm-up; a slowly increasing noise injection drives R downward, passing through the regime boundaries at 0.80, 0.50, and 0.30 (grey dashed lines), illustrating graceful multi-step degradation rather than abrupt failure. (B) Three-dimensional phase-difference matrix $|\phi_i - \phi_j| \bmod 2\pi$ at the end of a super-critical simulation ($K = 2.5 K_c$, $N = 20$). The surface is nearly flat with small corrugations, confirming that all agent pairs are tightly phase-locked with $|\phi_i - \phi_j| \ll 1$ rad. (C) Density-normalised ΔP histograms from four heterogeneous sensor types (temperature, pressure, light, vibration) plotted on a common axis. Each domain contributes a distinct but overlapping distribution, demonstrating that diverse physical modalities are composable in a unified ΔP space via the multi-domain transduction framework. (D) Running $R_{\text{ens}}(t)$ during a decoherence event (blue curve) with the detection threshold at $R = 0.50$ (red dashed line). The trajectory drops below threshold at a well-defined time (red dot), at which point the federation scheduler can trigger a re-synchronisation protocol; the detection latency is bounded by $m^* = \lceil 1/\varepsilon \rceil$ timing events.

mand injection into safety-critical sensors. Decoherence alerts serve as automatic fault indicators (Theorem 7.2).

Industrial IoT Mesh. TSF replaces 6TiSCH [22] scheduling overhead in dense IoT deployments. In phase-locked operation, no slot-scheduling messages are needed; each device emits pulses and reads cell assignments locally.

Satellite Constellation Coordination. Inter-satellite links with latency τ shift the effective coupling to $K_{\text{eff}} = K e^{-\omega\tau}$ (propagation delay reduces coherence). For low-earth orbit constellations with $\tau \approx 1$ ms, the effective coupling remains above K_c for crystal oscillators with $\sigma_\omega \leq 10$ ppb.

Autonomous Vehicle Platoons. Vehicle platooning requires $O(1)$ latency for braking consensus. In phase-locked TSF, the cell assignment for a “braking” event (detected via IMU deceleration spike shifting $\Delta P^{\text{(IMU)}}$) propagates to all vehicles within one timing event, satisfying hard real-time constraints.

Table 2: Coordination cost per decision round vs. swarm size n

Protocol	Cost (msgs)	Notes
TSF (Phase-Locked)	0	Theorem 5.3
TSF (Coherent)	$O(\log n)$	Leader election
TSF (Hierarchical)	$O(n \log n)$	Tree propagation
TSF (Turbulent)	$O(n^2)$	All-pairs
Raft [15]	$O(\log n)$	Log replication
Paxos [10]	$O(n)$	Multi-Paxos
Naïve broadcast	$O(n^2)$	All-to-all

11 Evaluation and Discussion

11.1 Coordination Cost Scaling

Table 2 summarizes the coordination cost scaling of TSF compared to classical protocols as a function of swarm size n .

11.2 Trajectory Distinguishability Scaling

The quantity $T(m, D) = D(D + 1)^{m-1}$ grows super-exponentially with D (polynomial base, exponential exponent). For fixed $m = 10$ and $D \in \{10, 30, 100\}$:

$$T(10, 10) \approx 2.6 \times 10^{10}, \quad T(10, 30) \approx 2.5 \times 10^{15}, \quad T(10, 100) \approx 1.1 \times 10^{20}$$

This demonstrates that heterogeneous sensor diversity compounds *multiplicatively*: each new sensor type added

Algorithm 6 Graceful Decoherence Handling

Input: Live R_{ens} stream; current regime r ; decoherence handler table $\mathcal{H}$

Output: Adapted coordination protocol

```
1:  $r_{\text{prev}} \leftarrow r$ 
2: loop
3:    $R_{\text{ens}} \leftarrow \text{PHASECOHERENCE}(\text{current phases}, W)$ 
4:    $r \leftarrow \text{CLASSIFYREGIME}(R_{\text{ens}})$ 
5:   if  $r < r_{\text{prev}}$  then  $\triangleright$  Decoherence event detected
6:      $m^* \leftarrow \lceil 1/\varepsilon \rceil$   $\triangleright$  Theorem 7.2
7:     Broadcast regime-transition alert (timing pulse only)
8:     Activate fallback protocol  $\mathcal{H}[r]$ :
9:       if  $r = 4$ : switch to consensus mode
10:      if  $r = 3$ : switch to hierarchical cascade
11:      if  $r = 2$ : switch to all-pairs constraint propagation
12:      if  $r = 1$ : enter emergency quorum mode
13:      Attempt re-entrainment: increase  $K$  by step  $\Delta K$ 
14:      Check coupling against  $K_c$   $\triangleright$  Algorithm 2
15:       $r_{\text{prev}} \leftarrow r$ 
16:      else if  $r > r_{\text{prev}}$  then  $\triangleright$  Re-entrainment detected
17:        Deactivate fallback; resume higher-regime protocol
18:         $r_{\text{prev}} \leftarrow r$ 
19:      end if
20:      await next timing event
21: end loop
```

to the federation exponentially increases the number of distinguishable joint swarm states.

11.3 Fault Tolerance

By Corollary 7.3, a single agent failure degrades the ensemble coherence by $\Delta R_{\text{ens}} \approx R_{\text{ens}}/n$ (removing one term from the sum in (2)). For $n = 20$ agents at $R_{\text{ens}} = 0.98$, removing one agent reduces R_{ens} to approximately $0.98 \cdot 19/20 = 0.931$, which remains within Regime 4 (Coherent). The swarm continues coordinating with $O(\log n)$ overhead, experiencing no catastrophic failure.

11.4 Limitations and Future Work

The current analysis assumes that the infrastructure coupling parameter K is controllable. In practice, K depends on physical channel characteristics (path loss, interference) and may not be adjustable without hardware changes. Future work will address: (1) adaptive coupling via variable-power pulse emission; (2) heterogeneous cou-

pling topologies beyond all-to-all (using graph Laplacian generalizations [14]); (3) time-varying frequency distributions (agents entering/leaving the federation dynamically); (4) experimental validation using MEMS oscillator hardware and sub-microsecond timestamping (IEEE 1588 [5]).

The thermodynamic limit assumption ($n \rightarrow \infty$) in Theorem 4.1 may not hold for small swarms ($n < 20$). For finite n , corrections of order $O(1/n)$ must be incorporated, which we defer to future analytical work.

12 Conclusion

We have introduced Temporal Swarm Federation, a coordination paradigm in which multi-agent systems achieve consensus through oscillator phase entrainment, with no semantic payload in any communicated signal. The central results are:

- (1) The ensemble’s coordination capability is completely characterized by the Kuramoto order parameter R_{ens} , which partitions the swarm into five regimes with analytically sharp phase-transition boundaries.
- (2) The critical coupling threshold $K_c = 2\sigma_\omega/\pi$ (Theorem 4.1) determines whether the swarm can sustain phase locking.
- (3) In the phase-locked regime ($R_{\text{ens}} \geq 0.95$), coordination overhead is provably zero (Theorem 5.3), analogous to superconductivity in condensed-matter physics.
- (4) Heterogeneous sensor fusion in a D -channel timing space yields $T(m, D) = D(D+1)^{m-1}$ distinguishable joint trajectories, growing super-exponentially with D (Theorem 6.1).
- (5) The absence of semantic payloads provides structural immunity to content injection attacks (Theorem 9.1), with bounded vulnerability to decoherence attacks (Proposition 9.3).

Temporal Swarm Federation establishes a new trade-space for multi-agent system design: when physical oscillators can be phase-entrained, content-bearing coordination protocols become unnecessary, and the resulting system is faster, more scalable, and structurally more secure. The paradigm is particularly promising for resource-constrained embedded swarms—drones, IoT mesh nodes, satellite constellations—where minimizing per-agent communication overhead is essential.

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